

Berkeley and Heiserman as an Unexhausted Architecture for Embodied Machine Intelligence

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Abstract

Edmund C. Berkeley is usually remembered as a writer who helped connect symbolic logic to computing machinery. That description is correct, but incomplete. Read across Berkeley’s machine-oriented writings and projects, the central concern is broader: to show that symbolic logic can serve as a practical design language for machines that acquire information, retain it, respond appropriately to changing conditions, and organize their behavior over time. This paper takes *Symbolic Logic and Intelligent Machines* as the principal text in that effort, while treating it as representative of a larger Berkeley program that links logical form, circuitry, control, and intelligent behavior. On this reading, Berkeley does not merely treat intelligence as abstract symbol manipulation. He repeatedly defines intelligent machines in operational terms, ties logic to hardware realization, and describes machine behavior through the coordinated interaction of inputs, outputs, memory, calculation, control, states, and events. His treatment of robots and machine activities accordingly moves beyond static logical form toward temporally extended, environment-coupled behavior. David L. Heiserman’s machine-intelligence work extends this program toward adaptive creature architectures built around memory, confidence, and generalization. Taken together, Berkeley and Heiserman can be read not as exhausted historical episodes but as contributors to a still under-tested architectural approach to embodied robotic cognition.

1 Introduction

Contemporary AI and robotics research often oscillates between large-scale statistical learning, abstract planning systems, and highly specialized control solutions. In that environment, behavior-organizing approaches rooted in explicit states, logical conditions, hardware-near control, and modular sensorimotor structure can be dismissed too quickly as dated. This paper begins from the opposite suspicion: that these design commitments remain fresh, technically coherent, and experimentally available for robotic cognition.

Edmund C. Berkeley is commonly placed in the history of symbolic logic, early computing, and machine thought. That placement is justified. *Symbolic Logic and Intelligent Machines* is overtly concerned with Boolean algebra, logical relations, class reasoning, and the design of information-handling machinery [Berkeley, 1959]. Yet this standard placement risks narrowing Berkeley’s significance. He was not only interested in the formal calculability of statements and classes. He was also interested in how reason-like competence could be built into machinery that senses, stores, decides, and acts.

The aim of this paper is to argue that Berkeley should be read as an architect of embodied machine intelligence rather than only as a symbolic logician of machines. He is not a full theorist of adaptive embodied AI in the Heiserman sense, but neither is he adequately understood as a purely formal or disembodied thinker. His work gives a logically rigorous account of how physically organized machines can take in information, retain it, operate on it, and guide their own activity over time [Berkeley, 1959, pp. 66–70]. When read alongside Heiserman’s creature-based machine intelligence, Berkeley becomes part of a still-open design program rather than a closed episode in the history of ideas.

2 Research Question and Thesis

The guiding question is how Berkeley’s *Symbolic Logic and Intelligent Machines* should be read today: merely as a period work of symbolic logic applied to machinery, or as a still-usable account of embodied, behavior-organizing intelligence.

The working thesis is that Berkeley’s work is best read as a technically viable account of embodied machine intelligence whose implications have not been exhausted. He inherits formal machinery from Boole, Venn, and Shannon, yet uses that inheritance to theorize intelligent systems as physically realized, temporally organized, and behaviorally active machines [Boole, 1847, 1854, Venn, 1894, Shannon, 1938]. Heiserman extends the same broad design space toward adaptive creature intelligence organized around memory, confidence, and generalization [Heiserman, 1981]. More strongly, Berkeley and Heiserman together can be read as defining an underused architectural alternative whose engineering potential has not been fully tested against current assumptions about how robotic intelligence must be built.

3 Literature Review

The literature relevant to this paper falls into five overlapping bodies of literature. The first and most important consists of the primary sources that directly carry the argument: Berkeley, Boole, Venn, Shannon, and Heiserman. Berkeley is the principal object of interpretation. Boole provides the originating algebraic logic that Berkeley explicitly treats as foundational [Boole, 1847, 1854]. Venn represents the strengthened class-logic and symbolic tradition that Berkeley inherits and occasionally names directly [Venn, 1894]. Shannon supplies the crucial bridge from logical relations to relay and switching hardware [Shannon, 1938]. Heiserman, by contrast, is not part of Berkeley’s own bibliographic world, but functions as a comparison point that clarifies how this same design space can be pushed toward adaptive and experiential machine intelligence [Heiserman, 1981].

The second body of literature consists of immediate contextual works in symbolic logic, especially Basson and O’Connor’s *Introduction to Symbolic Logic*. These are useful not because they define Berkeley’s project, but because they help establish what a more orthodox mid-century symbolic-logic presentation looked like [Basson and O’Connor, 1960]. Against that background,

Berkeley appears less like a standard textbook expositor and more like a writer pressing logic toward hardware, behavior, and machine organization.

The third body of literature is the cybernetic and control literature: Wiener, Ashby, Turing, and Walter [Wiener, 1948, Ashby, 1952, 1956, Turing, 1948, 1950, Walter, 1953]. These works are not needed to prove Berkeley’s core claims, but they help situate him within a broader postwar field concerned with control, communication, regulation, and autonomous machinery. Their function in this paper is therefore contextual and comparative, not genealogical.

The fourth body of literature is behavior-based or embodied-machine comparison, especially Braitenberg and Brooks [Braitenberg, 1984, Brooks, 1986, 1991]. These references are used cautiously. They do not show influence on Berkeley. Instead, they show that discussions of intelligent behavior can shift away from symbolic representation toward sensorimotor organization and situated control. Berkeley becomes especially useful in this frame because he retains logical structure while still treating machines as active, organized, and environment-facing systems.

The fifth and lightest body of literature is secondary historical or philosophical interpretation. This paper uses such literature only sparingly. Its method is primarily analytic and primary-source driven. The goal is not to subsume Berkeley under an already settled historiography, but to show from the texts themselves that his work occupies a more behavior-centered and embodied position than is usually granted.

It is also important to state the novelty claim carefully. We have not identified a prior literature that explicitly joins Berkeley and Heiserman in a single embodied-machine-intelligence architecture. The existing literature gives us Berkeley as a figure in symbolic logic and machine thought, Heiserman as an experimenter in adaptive robot intelligence, and recent work on state-based robotic behavior as a separate contemporary concern. What appears to be missing is a publication that explicitly argues that Berkeley and Heiserman belong to the same technically significant line of development. This paper proposes that linkage and argues for its engineering significance, while also clarifying its place in the literature.

4 From Published Work to Live Engineering Hypothesis

The argument of this paper is not that Berkeley and Heiserman merely deserve a more generous place in intellectual memory. The stronger claim is that they define a partially neglected engineering line. In Berkeley’s case, this line joins logical structure to hardware, control, state organization, and robot behavior. In Heiserman’s case, it extends into machine creatures that accumulate experience, retain successful responses, and later generalize from them under confidence-sensitive conditions [Heiserman, 1981, pp. 18–20].

What makes this line interesting now is not antiquarian recovery but architectural coherence. Both writers assume that intelligent machinery must be organized around explicit states, condition-sensitive transitions, sensorimotor coupling, and the structured coordination of internal and outward behavior. Even where the two differ, they differ along a productive axis: Berkeley provides

more formal and engineered control organization, while Heiserman pushes further toward adaptive behavior and memory-driven response.

This does not yet prove that their architecture offers the best path forward for modern robotics. It does, however, justify treating it as a live engineering hypothesis rather than a completed or obsolete episode. The question becomes whether behavior-organizing, state-based, hardware-near architectures remain powerful in domains where deterministic timing, embodied interaction, modular design, and constrained compute still matter.

5 Logic Embodied in Machinery

One of Berkeley's major contributions is to insist that symbolic logic is not merely a way of describing reasoning in abstraction. It is also a way of designing machinery. This is the decisive move from logic as formal analysis to logic as implemented intelligence. The bridge here is Shannon. Once Boolean relations can be realized in relay and switching circuits, conditions and inferences no longer remain on paper or in thought alone. They become structures of hardware [Shannon, 1938].

Berkeley adopts this possibility as fundamental. In his treatment, classes, statements, and logical conditions become design elements for circuits, control systems, and automatic machines. This means that logic is not simply explanatory; it is operational. Berkeley's interest in symbolic logic is therefore inseparable from his interest in machinery capable of carrying out organized, reason-like operations in the world [Berkeley, 1959].

6 Intelligent Machines as Organized Systems

The strongest evidence for an embodied reading appears in Berkeley's own definitions and examples. He lists a wide range of machines as "intelligent machines," including traffic-light controllers, air-traffic systems, factory control equipment, game-playing machines, reading and recognizing devices, robots, and digital computers [Berkeley, 1959, pp. 66–69]. These are not merely disembodied theorem provers. Many are directly engaged with environments, signals, movement, and control.

Berkeley then proposes a general architecture shared by almost every intelligent machine. Such a machine, he says, has inputs, outputs, storage or memory, calculating capacity, and a control unit [Berkeley, 1959, p. 70]. This matters conceptually because it frames intelligence not as a single faculty but as the coordination of sensing, acting, remembering, processing, and directing. Especially important is Berkeley's remark that the control unit may itself be subject to instructions calculated by the machine, allowing the machine in some sense to guide itself [Berkeley, 1959, p. 70]. This is modest autonomy, but it is autonomy nonetheless.

7 States, Events, and Temporally Extended Behavior

If Berkeley had stopped with Boolean algebra and machine architecture, the case for an embodied reading would remain suggestive. It becomes much stronger in his treatment of the algebra of states and events. There Berkeley explicitly argues that Boolean calculus must be extended in order to come “into closer relation with the real world,” since classes and statements change with time and since machine operation involves changing states and events rather than static truths alone [Berkeley, 1959, pp. 145–147].

A state lasts for a time; an event is brief. This distinction allows Berkeley to formalize persistence, interruption, transition, delay, and sequence. He also insists on the practical importance of time units, step-functions, and even intervals in which a machine state is temporarily undefined during transition [Berkeley, 1959, pp. 145–147]. The importance of this move is difficult to overstate. Intelligent behavior becomes describable as a structured passage through temporally extended conditions rather than as a mere arrangement of timeless logical forms. In this respect Berkeley moves from logic as static relation toward logic as an account of organized activity.

8 Robots and Behavioral Programs

Berkeley’s robot chapters make the behavioral implications even clearer. He defines a robot as a machine able to “behave” by itself, taking in sensations by means of sensing devices, performing actions by means of acting devices, and correlating sensations and actions by means of circuits or mechanisms expressing instructions [Berkeley, 1959, pp. 177–178]. This is already much closer to an agent-based conception of intelligence than to a purely symbolic one.

His examples are revealing. Berkeley’s electronic robot squirrel “Squee” is organized around activities such as hunting, homing, and depositing, each of which persists until relevant conditions terminate it [Berkeley, 1959, pp. 177–180]. Likewise, the “robot animal” problem is introduced explicitly as a programming problem in which one must assign a number of states to a robot and then specify changes from one state to another when certain events happen [Berkeley, 1959, pp. 177, 180–181]. These are not merely symbolic demonstrations. They are behavior-organizing schemes in which logical conditions select and govern activity.

9 Berkeley and Heiserman

The comparison with David Heiserman clarifies both Berkeley’s strengths and his limits. Heiserman’s *Robot Intelligence ... with Experiments* presents an evolutionary hierarchy in which Alpha-class creatures rely on constrained reflexive activity, Beta-class creatures add a memory of past successful responses, and Gamma-class creatures generalize from prior experience while building confidence in successful patterns of action [Heiserman, 1981, pp. 18–20]. In that framework, machine intelligence is lived by a creature in an environment. It senses, acts, remembers, adapts, and in the Gamma case anticipates.

Berkeley does not reach this level of adaptivity. His machines are mostly engineered top-down rather than self-modifying through reinforcement or confidence-based generalization. He does not provide Beta-style stored response schemes or Gamma-style conjectural extensions of prior success. Even so, the continuity remains important. Berkeley already supplies the control-and-behavior side of the story: intelligence as physically realized organization, behavior as transitions among activities, sensing and acting as core capacities, and internal control as the basis of outward competence.

The difference, then, is not that Berkeley lacks embodiment. The difference is that Berkeley’s embodiment is primarily engineered and prescribed, whereas Heiserman’s becomes increasingly adaptive and experiential. Berkeley links logic and circuitry to organized machine behavior; Heiserman pushes the same broad architectural space toward creature-like adaptation.

10 A Contemporary Use-Case

Recent work suggests that this state-centered and behavior-organizing perspective still has technical relevance. Christopher A. Tucker’s 2026 arXiv paper, *A Behavioral State Vocabulary in Sony ERS-111 R-CODE*, argues from a corpus-level analysis of AIBO behavior diagrams that many superficially different routines are structured by a compact embodied grammar organized around initialization, sensing, iterative action, synchronization, and recovery [Tucker, 2026]. Tucker further argues that this kind of state-based abstraction is useful as an intermediate representation for constructing new encapsulated behavior routines on constrained native robotic systems.

This is important for the present argument because it shows a contemporary use-case in which behavioral state vocabularies are not treated as relics. Instead, they are treated as practically useful abstractions for embodied control. Tucker does not cite Berkeley or Heiserman as direct sources in the available abstract material, and no strong claim of influence is warranted here. But the resonance is striking. What Berkeley formalized as states, events, control, and organized machine activity, and what Heiserman developed into adaptive creature architectures, reappears as a useful representational strategy in modern constrained robotics.

The result is not proof, but support. It suggests that the Berkeley–Heiserman line can be read as a still-open research program in which explicit behavior organization remains technically meaningful.

11 Conclusion

Berkeley should not be reduced to a symbolic logician who happened to mention machines. His work deserves to be read as an effort to show how intelligence can be built into physically organized systems that sense, remember, control, and act over time. His taxonomy of intelligent machines, his five-part architectural scheme, his extension of logic into states and events, and his robot examples together support a reading of Berkeley as a contributor to embodied machine intelligence in a

control-oriented sense.

This conclusion should be stated carefully. Berkeley does not formulate adaptive, reinforcement-based, or homeostatic machine intelligence in the Heiserman manner. But that limitation clarifies his importance rather than diminishing it. Berkeley shows how formal logical structure can govern temporally extended, physically embodied activity, while Heiserman shows how such organization can be pushed toward adaptive creature behavior.

The broader implication is that Berkeley and Heiserman together define more than a sequence of old publications. They define an underexplored engineering architecture: one in which logic, explicit state organization, embodied control, behavioral modularity, and adaptive response are treated as compatible ingredients of machine intelligence. Contemporary work on behavioral state vocabularies for constrained robotic systems lends support to taking that architecture seriously as a living research direction rather than as a closed curiosity [Tucker, 2026]. The proper conclusion, then, is not that this approach has already been vindicated, but that it deserves renewed and systematic experimental attention.

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